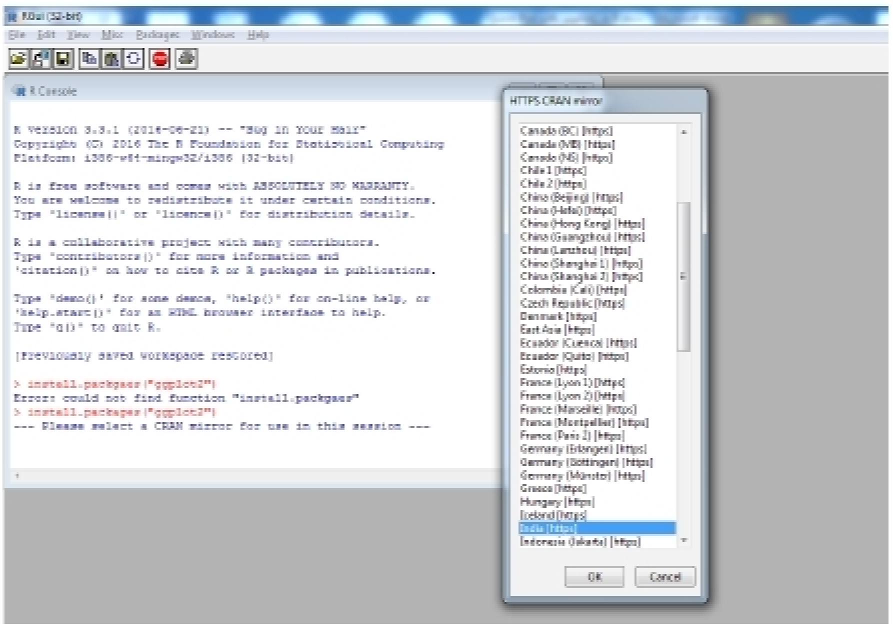

Figure 1: Installation of *ggplot2*

Now, you are able to perform the functions of ggplot2. The basic components of ggplot2 are:

- *ggplot()* function to create a new ggplot.
- *aes()* parameter to construct an aesthetic mapping of the plot. Each 'aesthetic' can be used in a number of ways based on the requirement of the plot.
- '+' component to add other components to a basic plot.
- *ggsave()* function to save the plots with default settings.
- *qplot()* or *quickplot()* function to generate basic plots quickly.

qplot() function with useful options

The *qplot()* function of the ggplot2 library is a powerful function to generate different types of plots quickly, and its general syntax is:

```
qplot(n1,n2,….,df,facets,geom,xlim,ylim,log,main,xlab,ylab,asp)
```

…where,

n1 n2,…, are the aesthetics passed into each layer. They are the values that we want in the graph;

df is the data frame, *tt* may be optional;

facet indicates the faceting formulae;

geom is the character vector that indicates the geom to draw. By default, it is a point (dot) if both values are specified or a histogram if there is only one value;

xlim and *ylim* are the axes limits;

log is a character vector indicating which axes values should be logged;

main, xlab and *ylab* are the character vectors to give the titles of three entities; and

asp indicates the y/x aspect ratio.

qplot() for different categories of graphs

The *qplot()* function is very vital for generating different categories of graphs like histograms, scatter plots, box plots, density plots, dot plots, violin plots, etc, quickly.

A histogram represents the distribution of numerical data.

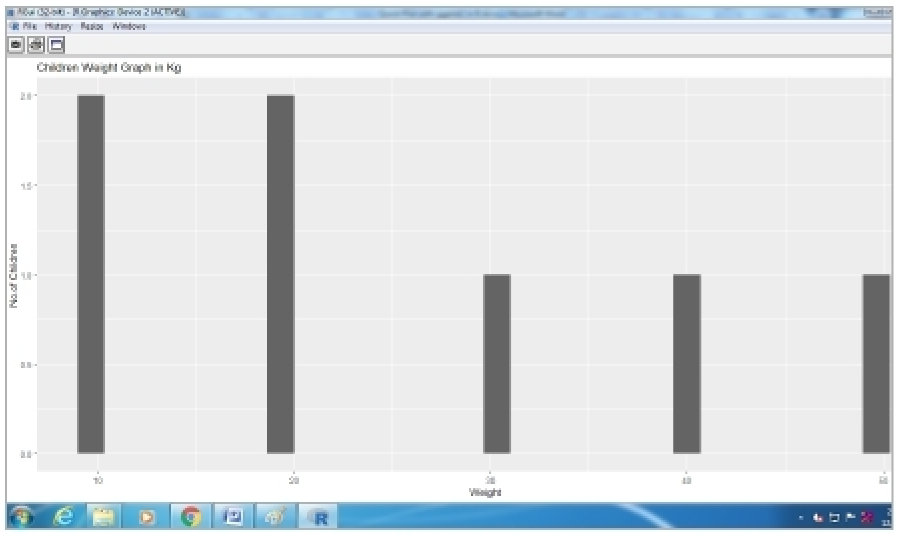

Figure 2: Histogram of children's weights using *qplot*

It relates to a single variable. Assume that the weight of several children is stored in one vector variable *x*.

```
x <- c(10,20,30,40,10,20,50)  // stores the weights of children
```

Now, we can generate the histogram for this using the *qplot()* function with the following command (Figure 2):

```
> qplot(x,main="Children Weight Graph in Kg",xlab="
Weight",ylab="No.of Children")
```

A scatter plot is a two-dimensional visualisation technique that uses the dots to represent the data. For example, vector *x* stores the weight and *y* stores the height of the children.

```
x <- c(10,20,30,40,10,20,50)  // Stores the weight of children
 y<-c(75,78,80,85,67,78)                        // Stores the
height of  children
```

The following *qplot()* function is used to represent the scatter plot of the above situation (Figure 3).

```
>qplot(x,y,geom=c("point","smooth"),main="Correlation of Weight
and
            Height",xlab="weight",ylab="height")
```

The boxplot is another technique for visualisation that

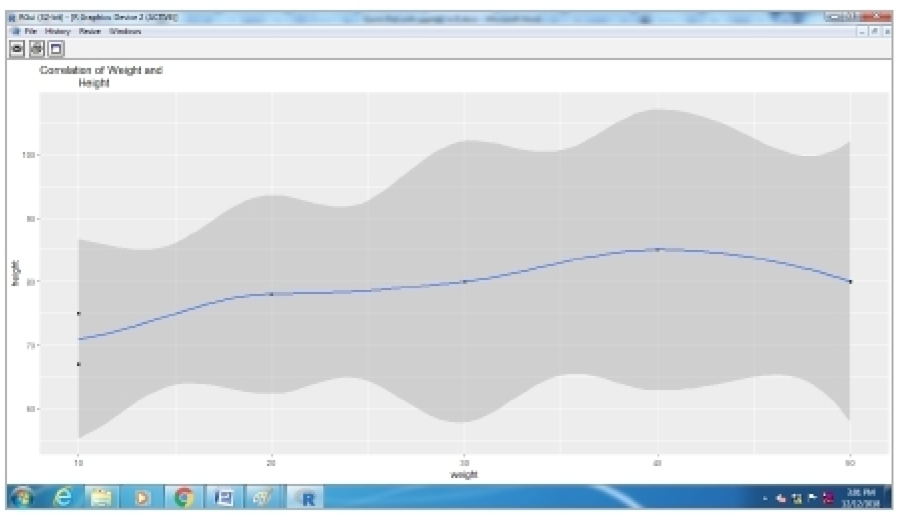

Figure 3: Scatter plot of correlation of weight and height using *qplot*